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Stepwise tracking strategy to screen ingredient from *Galla Chinensis* based on the "mass spectrometry guided preparative chromatography coupled with systems pharmacology"

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ABSTRACT

Ethnopharmacological relevance: Galla chinensis, a traditional Chinese herbal medicine, was widely used to treat ulcerative colitis (UC) in folk prescriptions, however, its active ingredients and mechanism of action in the treatment of UC remain unclear.

Aim of the study: The aim of our study was to discover the lead compounds and anti-inflammatory active ingredients of *Galla chinensis* and clarify their molecular mechanism for UC treatment.

Materials and methods: The ingredients of *Galla chinensis* were prepared by column and mass spectrometry guided preparative chromatography. Besides, the relationship among the ingredients of *Galla chinensis* and targets was predicted by systems pharmacology. Additionally, Lipopolysaccharide (LPS)-induced RAW264.7 macrophages were used as *in vitro* model. The cell viability, the level of the pro-inflammatory factors, the generation of reactive oxygen species (ROS), and trans epithelial electric resistance (TEER) values were detected to screen out the active ingredients of *Galla chinensis*. Moreover, 4% dextran sodium sulfate (DSS)-induced ulcerative colitis mice were used as the UC animal model. The disease activity index (DAI), pathological degree of colon tissue, activities of antioxidant-related enzymes and expression level of pro-inflammatory cytokines were performed to assess the anti-UC effects of the active ingredients. Meanwhile, the mRNA expression level of inflammatory factors and antioxidant related genes were analyzed by real-time quantitative polymerase chain reaction (Q-PCR). And the expression of nuclear factor erythroid-2 related factor 2 (Nrf2) pathway related proteins, intestinal mucosal proteins and nuclear factor kappa-B (NF-κB) pathway related proteins in colon tissues were analyzed by Western Blotting.

Results: Herein, a stepwise tracking strategy was adopted to screen out the anti-inflammatory active ingredients of *Galla Chinensis* based on "preparative chromatography pharmacology combined with mass spectrometry guidance and system". 11 categories of ingredients of *Galla chinensis* were prepared and ethyl gallate (EG) was screened out the lead compound and anti-inflammatory active ingredient of *Galla Chinensis* through *in silico*, *in vitro* and *in vivo* studies. In addition, EG had a significant therapeutic effect on ameliorating DSS-induced UC mice and protected intestinal mucosal integrity through Nrf2 and NF-κB signaling pathway.

Conclusion: Ethyl gallate was the lead compound and anti-inflammatory active ingredient in *Galla chinensis*. And it was discovered for the first time that EG could treat mice with ulcerative colitis. This research not only found the lead compound of *Galla Chinensis* for UC treatment and determined the possible mechanism, but also provided valuable references for finding lead compounds from natural products by systems pharmacology coupled with equivalent components group technology.

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Abbreviations

UC	ulcerative colitis
LPS	Lipopolysaccharide
TEER	trans epithelial electric resistance
DSS	dextran sodium sulfate
DAI	disease activity index
Q-PCR	real-time quantitative polymerase chain reaction
Nrf2	factor erythroid-2 related factor 2
NF-κB	nuclear factor kappa-B
EG	ethyl gallate
WHO	World Health Organization
TJs	tight junction proteins
ROS	reactive oxygen species
MS	mass spectrometry
UF-LC	ultrafiltration liquid chromatography
FBS	Fetal bovine serum
ELISA	Enzyme-linked immunosorbent assay
DMEM	Dulbecco's Modified Eagle's Medium
MTT	3-(4,5-Dimethylthiazol-2-yl)-2,5- diphenyl tetrazolium bromide;
5-ASA	5-Aminosalicylic acid
SVM	Support Vector Machine
C-T	compound-target
C-T-P	compound-target-pathway
KEGG	Kyoto Encyclopedia of Genes and Genomes
TNF-α	necrosis factor-α
IL-1β	Interleukin-1β
IL-6	Interleukin-6
NO	nitric oxide
H&E	hematoxylin and eosin staining
MDA	Malondialdehyde
SOD	Superoxide Dismutase
MPO	Myeloperoxidase
PVDF	polyvinylidene difluoride
m/z	mass-to-charge ratio
RT	the retention time
CAT	Catalase
SOD_2	superoxide dismutase2
NQO1	Quinone oxidoreductase1
CE	the crude extract of *Galla chinensis*
iNOS	inducible nitric oxide synthase
ZO1	Zona occludin-1

1. Introduction

Ulcerative colitis (UC) is a relapsing inflammatory bowel disorder of the colon that causes mucosal ulceration, chronic inflammation of intestinal mucosa with diarrhea, abdominal pain and rectal bleeding (Kobayashi et al., 2020). Recent population-based studies revealed that the prevalence of inflammatory bowel disease surpasses 0.3% in westernised countries of North America, Oceania, and Europe. Meanwhile, the incidence of inflammatory bowel disease has been accelerating in newly industrialised areas such as South America, Africa and Asia (Ng et al., 2017). The recurrence of UC brings a heavy economic and health burden to the health system and patients. At present, mesalazine is the main choice of treatment for mild to moderate UC and glucocorticoids are used to treat UC flares. Immunosuppressants and biologics are used in the moderate to severe stage. However, long-term use of mesalazine will lead to nausea, vomiting, and even hepatotoxicity. Although glucocorticoids like prednisone and hydrocortisone could regulate immune responses by interacting with glucocorticoid receptors in the nucleus, they may accompany some adverse effects including diabetes, osteoporosis and systemic edema. Additionally, patients with chronic use of monoclonal anti-tumor necrosis factor monoclonal antibodies such as Infliximab are more susceptible to hepatitis B or tuberculosis (Danese et al., 2020). Thus, it's imperative to discover new anti-UC drugs with less side effects and natural products have always been recognized as the most valuable source of bioactive compounds and drug leads.

Galla chinensis was the gall caused by aphids on the *Rhus* leaves of Anacardiaceae family and had been widely used as the traditional Chinese folk medicine (Tian et al., 2009a). In *Compendium of Materia Medica*, an ancient Chinese medical book, *Galla chinensis* was recorded for the treatment of lung deficiency, lung heat, phlegm cough, diarrhea, night sweat, bloody stool, hemorrhoid and traumatic hemorrhage, etc (Djakpo and Yao, 2010). Moreover, the mature fruits of this plant had long been used to treat dysentery and diarrhea, as well as other gastrointestinal disorders in Asia (Kouitcheu et al., 2013). Phytochemical studies revealed that *Galla chinensis* was rich in gallotannin, a type of hydrolysable tannin, which might represent at least 70% of its total weight (Djakpo and Yao, 2010). Our previous studies found that gallotannin played a pivotal role in UC treatment through inhibiting NF-κB signaling pathway (Yu et al., 2017; Bo et al., 2018). Intestinal mucosal collapse, a hypothesis put forward by Michael Eisenstein, indicated that disruption of the intestinal wall enable bacteria that are normally well tolerated (or even beneficial to health) in the lumen to trigger a poorly controlled immune response that causes chronic inflammation and tissue damage (Eisenstein, 2018). It has been reported that glallotannin of berries and pomegranate fruits were significantly upregulated the expression of tight junction proteins (TJs) leading to decreased gut permeability to microbial ligands through AhR-Nrf2 pathways (Singh et al., 2019). Reactive oxygen species (ROS) played an important role in the pathogenesis of a variety of acute and chronic inflammatory diseases (Asakura and Kitahora, 2018). A remarkable increased release of ROS from epithelial cells such as neutrophils, eosinophils, and monocytes which are an essential part of the mucosal immune system, may aggravate intestinal epithelial inflammation of UC (Maldonado-Contreras and Mccormick, 2011; Aviello and Knaus, 2017). However, the relationship among intestinal mucosal integrity, oxidative stress and inflammation has not been fully understood and glallotannin from *Galla chinensis* is a mixture containing a series of galloylglucopyronoses, which makes it difficult to discover bioactive ingredients from *Galla chinensis* and explore its potential mechanism against UC (Abu-Reidah et al., 2015).

Equivalent components group is to find one of the many ingredients of traditional Chinese medicine that can basically represent the original prescription. However, conventional drug leads screening through isolating, characterizing and testing each chemical compound from herbal extracts one by one is labor-intensive and time-consuming. Mass spectrometry (MS) is increasingly applied to screening and assessing bioactive compounds from herbal medicines because of its high throughput and high availability (Crozier, 2003). Song et al. (2014) proposed a strategy of ultrafiltration liquid chromatography (UF-LC)/MS to evaluate the binding of candidate compounds to protein targets and combined it with *in silico* molecular docking to predict binding sites. More importantly, they developed a bioactive equivalence oriented feedback screening method that integrates LC/MS analysis, bioactivity *in vitro* and *in vivo* assays, as well as metabolomics profiling (Li et al., 2016). Liu et al. (2014) discovered a novel neuroprotective compound from the fruit of *Alpinia oxyphylla* using LC/MS, bioactivity-oriented fractionation, and chemometrics.

Although gallnut medicine had been used for thousands of years and there were many studies on *Galla chinensis*, its active ingredient for the treatment of UC was still unclear. In the present study, a strategy of systems pharmacology and equivalent components group technology based on the MS guided preparative chromatography was proposed for screening the lead compound and active ingredients of *Galla chinensis*. Different fractions of *Galla chinensis* extract was first separated by column chromatography and each ingredient of *Galla chinensis* was prepared by MS guided preparative chromatography. After predicting potential protein targets with systems pharmacology and screening their biological activity with RAW264.7 cells, dextran sulfate sodium (DSS)-induced mouse model of UC was established to assess therapeutic effects of the candidate ingredient ethyl gallate (EG) and found that EG exerted protecting effects on intestinal mucosal through factor erythroid-2 related factor 2 (Nrf2) and nuclear factor kappa-B (NF-κB) signaling pathway.

2. Materials and methods

2.1. Materials

Herbal Medicines. *Galla chinensis* (*Rhus chinensis* Mill.) was purchased from Hanzhong, Shanxi, China in February 2018. The insect galls were identified by Professor Bo Han (School of Pharmacy, Shihezi University).

Chemicals. Lipopolysaccharide (LPS) was purchased from Sigma Chemical Co. (St Louis, Mo, USA). Fetal bovine serum (FBS) was produced by Biological Industries (Grand Island, NY, USA). Dulbecco's Modified Eagle's Medium (DMEM) was purchased from Gibco (Shanghai, China). 3-(4,5-Dimethylthiazol-2-yl)-2,5-diphenyl tetrazolium bromide (MTT), Trypsin-EDTA digestive juice and Penicillin-streptomycin solution were obtained from Solarbio (Beijing, China). 5-Aminosalicylic acid (5-ASA) was obtained from Aladdin Industrial Corporation (Shanghai, China). ROS assay kit was purchased from Jiancheng Biological Engineering Co. (Nanjing, China). Enzyme-linked immunosorbent assay (ELISA) kits were purchased from Enzyme linked Biological Technology Corporation (Shanghai, China). Goat anti-Mouse antibodies and β-Actinwere purchased from ZSGB-BIO (Beijing, China). Anti-Occludin antibody (EPR20992) and Anti-Claudin-1antibody ab15098 were purchased from Abcam (Cambridge, UK). IKKβ (D30C6) Rabbit mAb, Phospho-IKKα/β (Ser176/180) Antibody II, NF-κB p65 (D14E12) XP® Rabbit mAb, Phospho-NF-κB p65 (Ser536) Rabbit mAb, Keap1 (D6B12) Rabbit mAb, HO-1 (E9H3A) Rabbit mAb and Nrf2 (D1Z9C) XP® Rabbit mAb were purchased from Cell Signal Technology Inc. (Beverly, MA, USA).

Instruments. Waters XEVO TQD HPLC-MS system (LC-MS, Massachusetts, USA), sample manager (Waters 2767, The United States), binary gradient module (Waters 2545, The United States), UV/Vis detector (Waters 2489, The United States), 515 HPLC pump (Waters, The United States), QDa detector (Waters, The United States). FA2014B electronic balance (Shanghai precision science instruments limited), TDL80-2B desktop centrifuge (Shanghai Anting scientific instrument factory), Type 371 CO_2 cell incubator (Thermo Fisher Scientific company, USA), IX71 inverted microscope (Olympus company, Japan), HVE-50 Autoclave (Hirayama company, Japan), XMark microplate reader (Shanghai shuangxu electronics co.), Millipore Millicell-ERS cell resistance meter (Thermo Fisher Scientific company, USA), Flow Cytometer (Thermo Fisher Scientific, USA), TDL80-2B desktop centrifuge (Shanghai Anting scientific instrument factory). Q-PCR instrument (GIAGEN, Hilden Germany).

2.2. Preparation of ingredients in Galla chinensis

2.2.1. Extraction and preliminary separation of ingredients in Galla chinensis

The dried sample of *Galla chinensis* was pulverized to a uniform size in a mill and passed through a sieve (60-mesh). 100.0 g powders of *Galla chinensis* were accurately weighed, and then extracted with water (1:3, w/v) under reflux for 3 times for 2 h. The combined filtrates were concentrated under the reduced pressure to yield the extract 82.0 g. The separation of the crude extract of *Galla chinensis* (CE) was performed using Polyamide column chromatography. Briefly, the crude extract was slowly loaded into Polyamide column (30 cm × 2.5 cm inner diameter), which was previously equilibrated with water. The column was eluted by ethanol in a gradient (20%–50%∼70%). The elution volume of each concentration was 3 times column volume. Then each fraction was concentrated at 60 °C by a rotary evaporator and freeze-dried for later use.

2.2.2. LC-MS identified ingredients of Galla chinensis

Identification of ingredients in *Galla chinensis* according to the previous methods (Shangzhi et al., 2020). Analytical chromatographic analysis was performed with a Waters HPLC-MS system (Waters 2695). The 20 μL volume sample was eluted in a gradient elution process using A (0.2% Formic acid solution) and B (acetonitrile): 0–10 min (93%–80% A), 10–35 min (93%–80% A), 35–50 min (80%–70% A), 50–60 min (70%–0% A). The flow rate is 1 mL/min. Mass spectrometry was carried out by a Xevo TQD system equipped with an electrospray anion mode, and the parameters for the mode of MS were set as follows: source temperature: 150 °C, desolvation temperature: 250 °C, mass range: 100–1250 amu, capillary voltage 2.64 kV, cone voltage 50 V.

2.2.3. Preparation and analysis of ingredients by LC-MS

The preparation of ingredients was performed on a HPLC-MS preparative system. An X Bridge C18 column (5 μm, Φ19 × 150 mm, Waters) was used for chromatographic separation. The sample was eluted in the following gradient elution program: A (0.2% Formic acid solution) and B (acetonitrile): 0–4 min (93%–93% A), 4–22 min (93%–80% A), 22–35 min (80%–75% A), 35–60 min (75%–0% A), injection volume: 800 μL, flow rate: 17 mL/min. Electrospray (ESI) ion mode was used as negative. The MassLynx Mass Spectrometry Software was used for quantitative analysis using the peak area normalization method, and the ingredients were characterized by analytical HPLC-ESI-MS.

2.3. In silico studies: target prediction and systems pharmacology

The systems drug-target model, mainly based on Random Forest (RF) and Support Vector Machine (SVM) could integrate information from medicinal herbs, chemical compounds, targets, and pharmacological characteristics (Zhang et al., 2019). Herein, a *Galla chinensis* ingredients library was constructed according to drug-target interactions using the criteria of SVM score ≥0.8 and RF score ≥0.7. Furthermore, after correlating the potential targets to related diseases, which were obtained from the PharmGKB data-base (Kraja et al., 2011), and Therapeutic Targets Database (http://bidd.nus.edu.sg). The pathway information was obtained by uploading the target data to the Kyoto Encyclopedia of Genes and Genomes (KEGG, http://www.kegg.jp) database. Networks of compound-target (C-T) and compound-target-pathway (C-T-P) in *Galla chinensis* were constructed by Cytoscape 3.2.1 software for subsequent analysis.

2.4. In vitro studies: screening active ingredients on RAW264.7 cells

Cell Culture. Human Caco-2 cells and mouse macrophage RAW264.7 were obtained from the cell bank of Shanghai Institutes for Biological Sciences. Caco-2 cells and RAW 264.7 cells were cultured in DMEM medium containing 10% FBS and 1% penicillin-streptomycin solution at

37 °C with 5% CO_2.

2.4.1. Cell viability assay

RAW264.7 cells were seeded at a density of 5×10^4 cells/well in 96-well plates overnight, and pre-treated with different concentrations of each ingredient (3.125, 6.25, 12.5, 25 and 50 μg/mL) or dexamethasone (positive control) for 1 h before stimulation with LPS (0.5 μg/mL). RAW264.7 cells viability was measured after 24 h of exposure to the tested each ingredient with a colorimetric assay based on the ability of mitochondria in viable cells to reduce MTT. The MTT solution was added at a concentration of 0.5 mg/mL into each well. After 4 h of incubation at 37 °C, the medium was discarded and the formazan blue formed in the cells was dissolved in DMSO. Optical density at 570 nm was determined with a microplate reader.

2.4.2. Measurement of tumor necrosis factor-α (TNF-α), Interleukin-1β (IL-1β), Interleukin-6 (IL-6) and nitric oxide (NO)

RAW264.7 cells (2×10^5 cells in 24-well plates) were treated with each ingredient (3.125, 6.25, 12.5 and 25 μg/mL) or dexamethasone (positive control) for 1 h, followed by LPS (0.5 μg/mL) for 24 h. Serum from mouse homogenate and supernatant after cell administration were taken. The levels of TNF-α, IL-1β, IL-6 and NO in the medium of cultured RAW 264.7 cells were measured by ELISA. The levels of TNF-α, IL-1β, IL-6 in the serum of mice were measured by ELISA. A450 nm (reference absorbance) was measured on a Model 680 Microplate Reader (Multiskan FC, Thermo).

2.4.3. Determination of intracellular ROS

Intracellular oxidative stress was detected using DCFH-DA as described by Choi YK et al. (Choi et al., 2018). After inoculating in a 6-well plate (4×10^5 cells/well) and incubating to reach 80% confluence, RAW264.7 cells were treated with each ingredient (6.25, 12.5 and 25 μg/mL) or dexamethasone (12.5 μg/mL, positive control) for 1 h, followed by LPS (0.5 μg/mL) stimulation for 24 h. Then the cells were washed twice with phosphate buffer solution (PBS) and incubated with 10 μM DCFH-DA in the dark for 40 min. The harvested cells were immediately analyzed by the flow cytometry.

2.4.4. Measurement of trans epithelial electric resistance (TEER) values

Caco-2 cells were cultured for 11 days to form a cell monolayer after seeding in 24-transwell cell culture chambers (6.5 mm diameter inserts, 3.0 μm pore size) (Corning Costar, Corning Incorporated, USA). Then RAW246.7 cells were seeded at the bottom of the well. After 3 days, RAW246.7 cells were pre-treated with each ingredient (6.25, 12.5 and 25 μg/mL) or dexamethasone (12.5 μg/mL, positive control) for 1 h and then stimulated with LPS (0.5 μg/mL) for 24 h. The complete medium was replaced with serum free medium and incubated for 20 min, and then the TEER values were measured using a resistor. The TEER values (T_V) is calculated by formulas (1):

$$TV = \frac{(T - TB)}{Membrane\ area^2} (\Omega / cm^2) \tag{1}$$

Where T is the resistance of the cell monolayer along with the filter membrane, T_B is the resistance of the filter membrane (black), and Membrane area = 0.332 cm^2.

2.5. In vivo studies: screening active ingredients on DSS-induced UC

DSS-induced Mouse Model of UC. Kunming mice (clean grade, half females and half males, 21–23 g) were provided by the Experimental Animal Research Center in Xinjiang (SCXK (xin) 2014-0001). They were adapted to the environment for a week and fed with standard pellets before the experiment. Mice were randomly assigned to 7 groups. The mice in the control group drank distilled water without DSS. The mice in the DSS group, the 5-ASA-treated group (100 mg/kg), CE-treated group (250 mg/kg), and EG-treated group (50, 150 and 250 mg/kg) drank 4% DSS freely each day. In the morning, enemas (enema tubes inserted 4 cm into the rectum) are performed in mice, and for 10 consecutive days, while the control group was administered saline solution as well as the DSS group in the same manner. Body weight, stool character and fecal occult blood of mice were monitored and calculated through the whole experiment (Philip et al., 2009).

2.5.1. Disease activity index (DAI)

DAI was calculated according to the scoring system as presented in Table S1. Weight losses of 0, 1–5%, 5–10%, 10–15% and >15% corresponded to the score of 0, 1, 2, 3, and 4. Stool consistency was graded from 0 to 4 (0 for normal well-formed particles; 1 for loose stools; 2 for semiformed stools; 3 for liquid stools; 4 for diarrhea). As to bleeding, 0, 1, 2, 3, and 4 stand for no blood, trace blood, mild hemoccult, obvious hemoccult, and gross bleeding, respectively. The sum of these subscores was designated as the DAI score.

2.5.2. Histopathological assay

Segments of colons were stained with hematoxylin and eosin staining (H&E) for light microscopic examination after fixing in 10% formaldehyde solution, embedding in paraffin and sectioning to 4 μm thickness. Histological scores were calculated according to a modified score system, which incorporates degree of inflammation (scale of 0–2), depth of inflammation (0–3), crypt damage (0–4) and percentage of area affected by inflammation (0–4) (Dieleman et al., 1998). The total score is from 0 (normal) to 13 (severe colitis).

2.5.3. Measurement of malondialdehyde (MDA), superoxide dismutase (SOD) and myeloperoxidase (MPO) level

For MDA and SOD determination, colon samples were excised and homogenized immediately at 4 °C. The supernatant was then collected and stored at −80 °C until use. Protein Quantification Kit was used to determine protein concentration and the level of MDA and SOD were measured using MDA and SOD kits in accordance with the manufacturer's instructions. The absorbance for MDA and SOD detection were 532 nm and 550 nm. The concentration of MDA and SOD in colon were expressed in nmol/mg and U/mg protein, respectively. For MPO measurement, colon tissues were homogenized in 1:19 (w/v) of buffer solution and the MPO activity was determined according to the instructions of test kits. Absorbance was set at 450 nm and the activity of MPO was defined as pg/mL protein.

2.6. Quantitative reverse transcription-polymerase chain reaction (Q-PCR) analysis

Extraction of total RNA from the mouse colon samples was performed according to the manufacturers procedure of TRIzol reagent (15596026, Thermo Fisher Scientific, USA). Reverse-transcription of the RNA into complementary DNA (cDNA) was realized in a RevertAid First Strand cDNA Synthesis Kit (K1622, Thermo Fisher Scientific, USA). QuantiNova™ SYBR® Green PCR kit (208054, QIAGEN Gmbh, Germany) was used to detect the mRNA expression. The sequences of primers are listed in Table 1 β-Actin was used as the reference gene.

2.7. Western blotting analysis

The colon tissues of each group were homogenized in the protein lysis buffer and protein was extracted. The appropriate concentration of sodium dodecyl sulfate-polyacrylamide gel was selected for protein separation according to different molecular weights. Target proteins were transferred to polyvinylidenedifluoride (PVDF) membranes after electrophoresis and the membranes were blocked with 5% nonfat milk for 1 h, followed by the incubation with primary antibodies against p65 (1:1000), p-p65 (1:1000), IKKβ (1:1000), p-IKKβ (1:1000), Keap1 (1:1000), HO-1 (1:1000), Nrf2 (1:1000) and β-actin (1:2000) overnight

Table 1
Primers used for Q-PCR.

Target genes	Primer sequence (5′–3')
iNOS	F: TTTGGAGCAGAAGTGCAAAGTCTC R: GATCAGGAGGGATTTCAAAGACCT
TNF-α	F: CCCTCACACTCAGATCATCTTCTC R: GGCTACAGGCTTGTCACTCG
IL-6	F: AGAAGGAGTGGCTAAGGACCAA R: ACGCACTAGGTTTGCCGAGTA
IL-1β	F: GCAACTGTTCCTGAACTCAACT R: ATCTTTTGGGGTCCGTCAACT
HO-1	F: GGCAGAGGGTGATAGAAGAGG R: AGCTCCTGCAACTCCTCAAA
NQO1	F: CGCGACTCCCACAAGGTT R: GTCCGACTCCACCACCTCC
SOD_2	F: AGCTGCACCACAGCAAGCAC R: TCCACCACCCTTAGGGCTCA
CAT	F: TCCGGGATCTTTTTAACGCCATTG R: TCGAGCACGGTAGGGACAGTTCAC
β-Actin	F: GGCGGACTGTTACTGAGCTG R: CTGCGCAAGTTAGGTTTTGTCA

Notes: F forward, R reverse.

at 4 °C. The next day, secondary antibodies, goat anti-rabbit or anti-mouse IgG (1: 10000) were incubated with the membranes at room temperature for 1 h. β-actin was used as an internal reference and the band density was quantified as the ratio to β-actin.

2.8. Statistical analysis

Significant differences ($P < 0.05$, $P < 0.01$ or $P < 0.001$) between the groups were determined using one-way analysis of variance (ANOVA) with SPSS software version 16.0. The data were expressed as mean ± SD.

3. Results

3.1. Identification of ingredients in Galla chinensis

Three fractions were obtained by ethanol gradient elution with polyamide as filler, in order to improve the efficiency of preparing ingredients of *Galla chinensis* by liquid chromatography-mass spectrometry. The mass spectrogram of each fraction was shown in Fig. 1. The fraction of eluted with 20% ethanol (Fr.A) contained the ingredients with *m/z* values of 331.20 and 169.07. The fraction of eluted with 50% ethanol (Fr.B) involved the ingredients with *m/z* values of 483.26, 321.09 and 635.34. The fraction of eluted with 70% ethanol (Fr.C) had

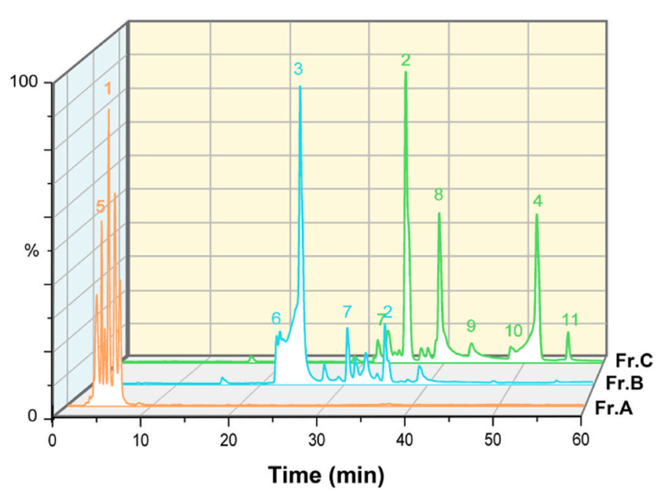

Fig. 1. Mass spectrometry analysis of each fraction of Galla chinensis. The fraction A was eluted with 20% ethanol. The fraction B was eluted with 50% ethanol. The fraction C was eluted with 70% ethanol.

the ingredients with *m/z* values of 197.07, 787.24, 939.27, 349.10, 1090.63 and 1243.30. Each ingredient of *Galla chinensis* was prepared from these fractions. The mass spectrogram of each ingredient and CE were shown in Fig. 2. The mass-to-charge ratio of these ingredients were *m/z* 331.20, *m/z* 169.07, *m/z* 483.26, *m/z* 321.09, *m/z* 635.34, *m/z* 197.07, *m/z* 787.24, *m/z* 939.27, *m/z* 349.10, *m/z* 1090.63 and *m/z* 1243.30. From our previously published studies and relevant literature analysis (Ren et al., 2021; Shangzhi et al., 2020; Tian et al., 2009b), the structure of these ingredients were identified as mono--*O*-galloyl-β-D-glucose, gallic acid, di-*O*-galloyl-β-D-glucose, digallic acid, tri-*O*-galloyl-β-D-glucose, EG, tetra-*O*-galloyl-β- D-glucose, penta-*O*-galloyl-β-D-glucose, ethyl digallate, hexa-*O*-galloyl-β-D-glucose and hept-*O*-galloyl-β-D-glucose. The retention time of EG prepared by us was consistent with that of EG standard.

The impurity contents in each ingredient were investigated by the peak area normalization method. Detailed information of each ingredient was listed in Table 2, such as the identified chemical names, mass-to-charge ratio (*m/z*), the retention time (RT), formula, concentration and peak area ratio (%).

3.2. Analysis of systems pharmacology

A C-T network depended on the candidate compounds and potential targets of *Galla chinensis* was constructed. The C-T network as shown in Fig. S1, including 41 nodes (11 categories of ingredients and 30 potential targets) and 93 sides. The average number of targets in each ingredient was 2.7, indicating that most of the ingredients regulated multiple targets to exert various therapeutic effects. Among 11 categories of ingredients, EG (M2), di-*O*-galloyl-β-D-glucose (M6) and tri-*O*-galloyl-β-D-glucose (M7) acted on 20, 12, and 16 targets, respectively, meaning they are crucial active ingredients of *Galla chinensis* because of their high degree of distributions in this network. Additionally, the results showed that some targets were attacked by multiple ingredients. Specifically, NF-κB1 was targeted by 4 ingredients, IL-1β, IL-6 and TNF were all targeted by 3 ingredients. Studies show that these four target points play a central role in regulating inflammatory responses (Neurath, 2014). Besides, 3 ingredients were targeted with Catalase (CAT) and superoxide dismutase 2 (SOD_2), which prevented excessive levels of ROS in cells (Patlevič et al., 2016).

The compound-target-pathway (C-T-P) network was constructed using the corresponding UC targets from KEGG database as bait to clarify the mechanism of active ingredients in treating UC. As shown in Fig. 3, 11 categories of ingredients, 30 targets and 41 pathways were used to generate the C-T-P network, consisting of 81 nodes and 233 sides. Each ingredient of *Galla chinensis* corresponded to multiple targets and multiple pathways. Especially, EG (M2), di-*O*-galloyl-β-D-glucose (M6) and tri-*O*-galloyl-β-D-glucose (M7) acted on a wealth of targets such as TNF, SOD_2, CAT, Quinone oxidoreductase 1(NQO1), IL-6 and IL-1β were closely related to the UC process. These targets corresponded to 6 pathways, *i.e.*, MAPK, PI3K-Akt, NOD-like receptor, TLR-like receptor, TNF, NF-κB and Nrf2-Keap1 signaling pathways which played an important role in inflammatory diseases. Among these, specifically, NF-κB and Nrf2-Keap1 signaling pathways played the highest role.

According to the predicted results of systems pharmacology, we speculated that the ingredients of *Galla chinensis* might reduce inflammation and oxidative stress injury by regulating NF-κB and Nrf2-Keap1 signaling pathways, thus reducing the severity of UC. Next, the predicted results were verified *in vitro* and *in vivo*.

3.3. Effects of each ingredient on cell viability

Cell viability was analyzed to screen the low toxic ingredients of *Galla chinensis*. The results were shown in Fig. 4, the cell viability was unaffected by LPS and the results were presented as relative cell viability referred to control (equal to 100%). The safe concentration (the cell survival rate is more than 80%) of the ingredients of *m/z* values 321.01

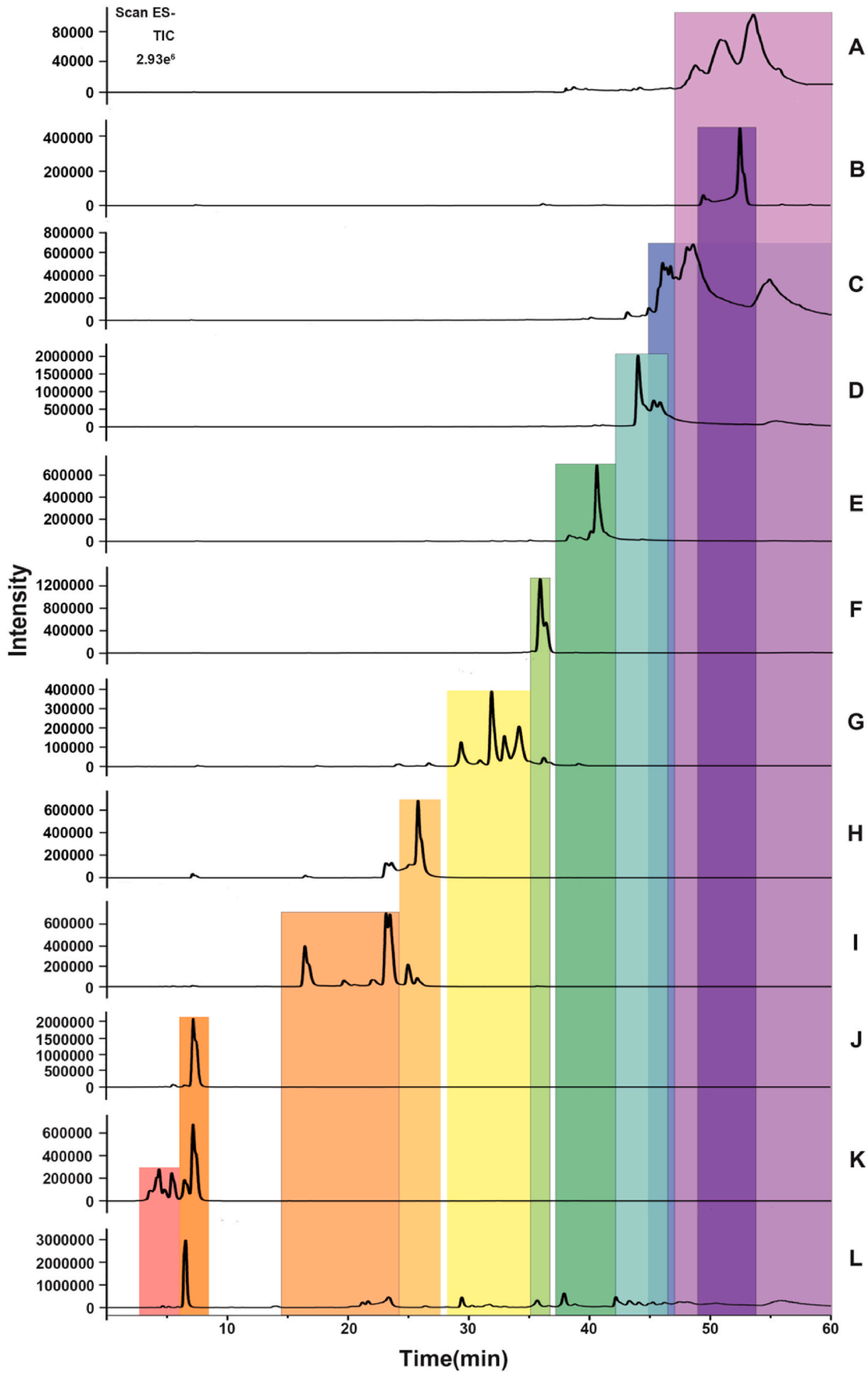

Fig. 2. Mass spectrometry analysis of each ingredient and CE. A to K were molecular distributions of *m/z* 1243.30, 349.10, 1091.86, 939.17, 787.19, 197.07, 635.30, 321.01, 483.22, 169.02 and 331.05 respectively. L is the total ion diagram of CE.

Table 2
Detailed information of each ingredient in *Galla chinensis*.

No.	The identified chemical names	*m/z*	RT (min)	Formua	Concentration (mg/mL)	peak area ratio (%)
1	gallic acid	169.0205	6.53	$C_7H_6O_5$	1	92.07
2	ethyl gallate	197.0735	35.83, 35.96, 36.31	$C_9H_{10}O_5$	1.2	95.15
3	digallic acid	321.1016	25.79, 26.19	$C_{14}H_{10}O_9$	1.2	90.88
4	ethyl digallate	349.1047	52.52	$C_{16}H_{15}O_{10}$	0.6	95.80
5	mono-*O*-galloyl-β-D-glucose	331.0498	2.45, 3.83, 6.42, 6.67	$C_{13}H_{16}O_{10}$	2	85.96
6	di-*O*-galloyl-β-D-glucose	483.2170	16.50, 23.75	$C_{20}H_{19}O_{14}$	1.4	97.45
7	tri-*O*-galloyl-β-D-glucose	635.2994	26.33, 29.46, 33.07	$C_{27}H_{23}O_{18}$	0.5	91.08
8	tetra-*O*-galloyl-β-D-glucose	787.1998	40.56, 40.60	$C_{34}H_{27}O_{22}$	1.4	92.23
9	penta-*O*-galloyl-β-D-glucose	939.1669	44.08	$C_{41}H_{31}O_{26}$	0.6	90.20
10	hexa-*O*-galloyl-β-D-glucose	1091.8642	47.14, 55.29	$C_{48}H_{35}O_{30}$	0.4	91.51
11	hept-*O*-galloyl-β-D-glucose	1243.3025	50.86, 55.22	$C_{55}H_{39}O_{34}$	0.4	94.22

Notes: Peak area normalization, as a rough method to investigate the purity of test product, calculates the peak area and sum of each ingredient, and then we calculated the percentage of the total peak area.

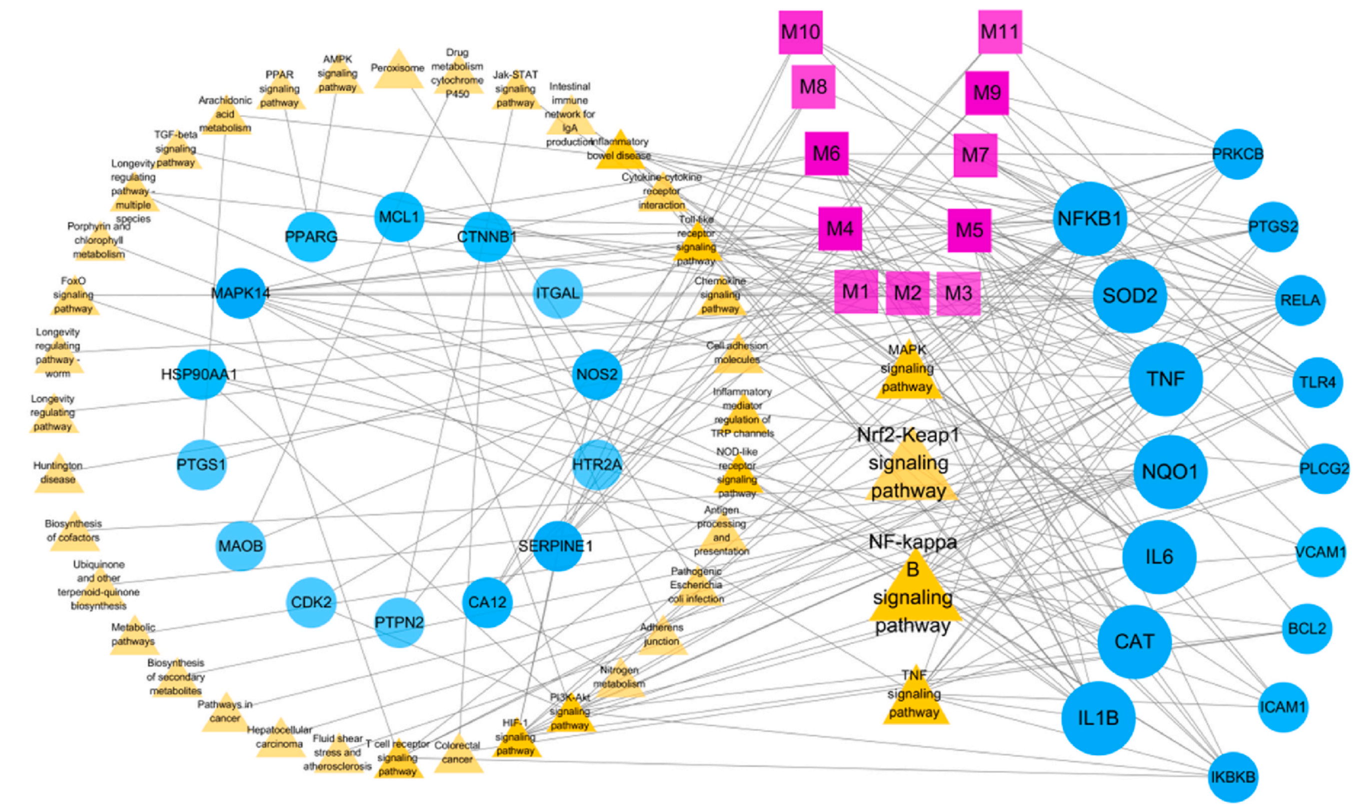

Fig. 3. C-T-P network of *Galla chinensis*. The purple rectangles represented the chemical ingredients of *Galla chinensis*, where M1 was gallic acid, M2 was ethyl gallate, M3 was digallic acid, M4 was ethyl digallate, M5 was mono-*O*-galloyl-β-D-glucose, M6 was di-*O*-galloyl-β-D-glucose, M7 was tri-*O*-galloyl-β-D-glucose, M8 was tetra-*O*-galloyl-β-D- glucose, M9 was penta-*O*-galloyl-β-D-glucose, M10 was hexa-*O*-galloyl-β-D-glucose, M11 was hept-*O*-galloyl-β-D-glucose. The blue circles represent targets related to the ingredients. The yellow triangles indicated the pathway of UC. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

(digallic acid), 331.05(mono-*O*-galloyl-β-D-glucose), 483.22 (di-*O*-galloyl-β-D-glucose), 635.30 (tri-*O*-galloyl-β-D-glucose), 197.07 (EG) and 787.19 (tetra-*O*-galloyl-β-D-glucose) was 25 μg/mL. The safe concentration of the ingredients with *m/z* values of 349.10 (ethyl digallate) and 939.17 (penta-*O*-galloyl-β-D-glucose) was 6.25 μg/mL. The safe concentration of ingredients with *m/z* values of 169.02 (gallic acid) and 1243.30 (hept-*O*-galloyl-β-D-glucose) was 3.125 μg/mL. The ingredient with *m/z* values of 1090.63(hexa-*O*-galloyl-β-D-glucose) had been showed toxicity at 3.125 μg/mL. Thus, 6 categories of ingredients of *Galla chinensis*, with *m/z* values of 197.07 (EG), 321.01 (digallic acid), 331.05 (mono-*O*-galloyl-β-D-glucose), 483.22 (di-*O*-galloyl-β-D-glucose), 635.30 (tri-*O*-galloyl-β-D-glucose) and 787.19 (tetra-*O*-galloyl-β-D-glucose) were chosen to perform subsequent experiments with the concentration doses of 3.125, 6.25, 12.5 and 25 μg/mL.

3.4. Inhibitory effect of ingredients on NO and cytokines production in LPS-stimulated RAW264.7 cells

To investigate whether 6 categories of ingredients of *Galla chinensis* affect the production of NO by LPS-stimulated macrophages, RAW264.7 cells were incubated with different concentrations of the ingredients for 1 h and then treated with 0.5 μg/mL LPS for 24 h. The levels of NO in the culture medium were measured by ELISA. As shown in Fig. 5A, LPS stimulation significantly increased the release of NO compared to the blank group ($P < 0.001$). In contrast, the ingredients with *m/z* values of 197.07 (EG), 483.22 (di-*O*-galloyl-β-D-glucose) and 635.30 (tri-*O*-galloyl-β-D-glucose) significantly inhibited the production of NO by LPS-activated RAW264.7 cells with a dose-dependent manner ($P < 0.01$). As shown in Fig. S2A, ingredients with *m/z* values of 321.01 (digallic

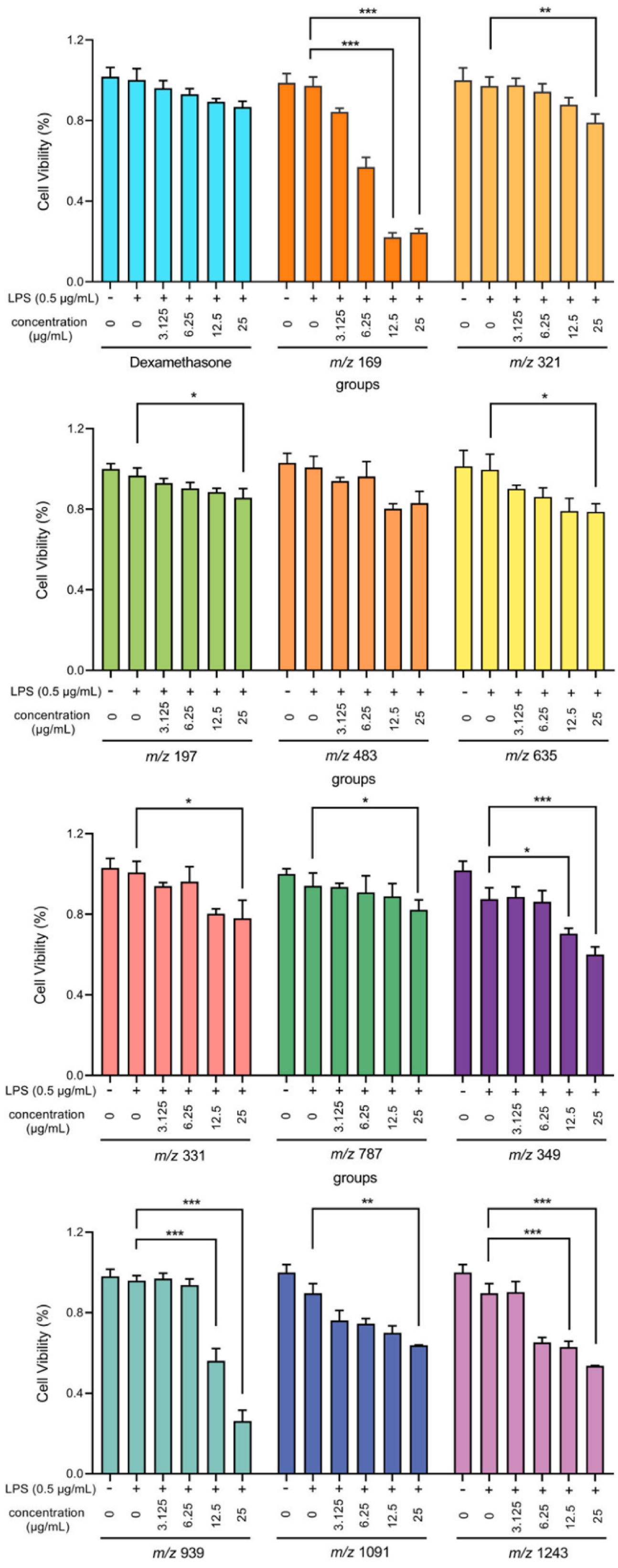

Fig. 4. Effects of the ingredients of *Galla chinensis* on the cell viability. Cells were incubated with different concentrations of the ingredients for 1 h and then treated with 0.5 μg/mL LPS for 24 h. The cell viability was determined by the MTT assay. The values are expressed as mean ± SD, n = 6. Values of $^{\#}P < 0.05$, $^{\#\#}P < 0.01$ and $^{\#\#\#}P < 0.001$ *vs.* the blank group. Values of $^{*}P < 0.05$, $^{**}P < 0.01$ and $^{***}P < 0.001$ *vs.* the model group.

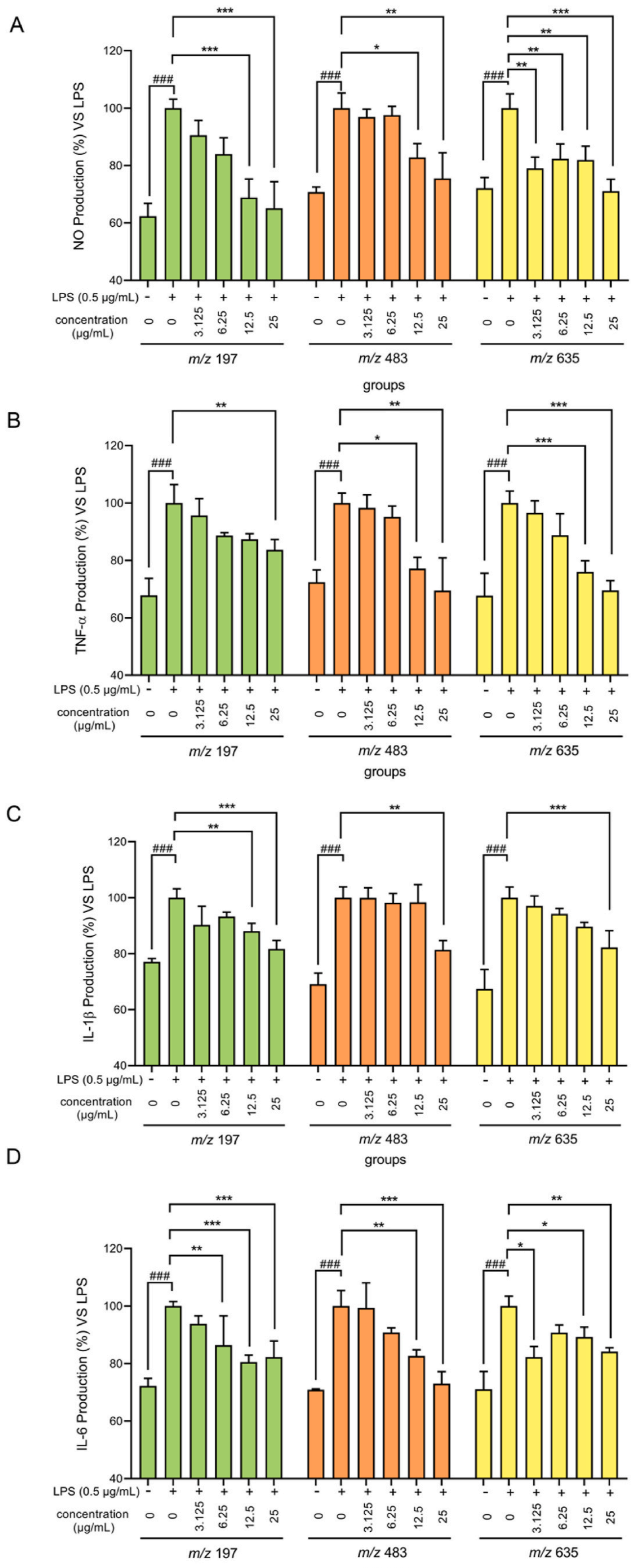

Fig. 5. Effects of the ingredients of *Galla chinensis* on the production of NO and cytokines in LPS-activated RAW 264.7 Cells. Cells were incubated with different concentrations of the ingredients for 1 h and then treated with 0.5 μg/mL LPS for 24 h. The levels of (A) NO, (B) TNF-α, (C) IL-1β and (D) IL-6 in the culture medium were measured by ELISA. The values are expressed as mean ± SD, n = 6. Values of $^{\#}P < 0.05$, $^{\#\#}P < 0.01$ and $^{\#\#\#}P < 0.001$ *vs.* the blank group. Values of $^{*}P < 0.05$, $^{**}P < 0.01$ and $^{***}P < 0.001$ *vs.* the model group.

acid), 331.05 (mono-*O*-galloyl-β-D- glucose) significantly inhibited the production of NO in LPS-activated RAW264.7 cells ($P < 0.01$), while the ingredient with *m/z* values of 787.19 (tetra-*O*-galloyl-β-D-glucose) inhibited the production of NO at dose of 25 μg/mL ($P < 0.05$). In summary, the ingredients with *m/z* values of 197.07 (EG), 321.01 (digallic acid), 331.05 (mono-*O*-galloyl-β-D-glucose), 483.22 (di-*O*-galloyl-β-D-glucose), 635.30 (tri-*O*-galloyl-β-D-glucose) and 787.19 (tetra-*O*-galloyl-β- D-glucose) had inhibitory effects on NO production.

In this study, the cytokine levels in the supernatant of LPS-stimulated RAW264.7 cells cultured with various concentrations of the ingredients were measured by ELISA. The experimental results showed that LPS stimulation significantly increased the release of TNF-α compared to the blank group ($P < 0.001$), while the ingredients with *m/z* values of 197.07 (EG), 483.22 (di-*O*-galloyl-β-D-glucose) and 635.30 (tri-*O*-galloyl-β-D-glucose) significantly inhibited the production of TNF-α in LPS-activated RAW264.7 cells ($P < 0.01$) (Fig. 5B). As shown in Fig. S2B, the ingredients with *m/z* values of 331.05 (mono-*O*-galloyl-β-D-glucose) inhibited TNF-α produced in LPS-activated RAW264.7 cells ($P < 0.05$), while the ingredients with *m/z* values of 321.01 (digallic acid) and 787.19 (tetra-*O*-galloyl-β-D-glucose) didn't significantly inhibit the production of TNF-α.

As shown in Fig. 5C, LPS stimulation significantly increased the release of IL-1β compared to the blank group ($P < 0.001$). In contrast, the ingredients with *m/z* values of 197.07 (EG), 483.22 (di-*O*-galloyl-β-D-glucose) and 635.30 (tri-*O*-galloyl-β-D-glucose) significantly inhibited the production of IL-1β in LPS-activated RAW264.7 cells ($P < 0.01$). As shown in Fig. S2C, the ingredients with *m/z* values of 321.01 (digallic acid) and 787.19 (tetra-*O*-galloyl-β-D-glucose) inhibited the production of IL-1β in LPS-activated RAW264.7 cells ($P < 0.05$), while the ingredient with *m/z* values of 331.05 (mono-*O*-galloyl-β-D-glucose) didn't significantly inhibit the production of IL-1β.

The results showed that LPS stimulation significantly increased the release of IL-6 compared to the blank group ($P < 0.001$), while the ingredients with *m/z* values of 197.07 (EG), 483.22 (di-*O*-galloyl-β-D-glucose) and 635.30 (tri-*O*-galloyl-β-D-glucose) significantly inhibited the production of IL-6 in LPS-activated RAW264.7 cells with a dose-dependent manner ($P < 0.01$) (Fig. 5D). As shown in Fig. S2D, the ingredients of *m/z* values 331.05 (mono-*O*-galloyl-β-D-glucose) inhibited the production of IL-6 in LPS-activated RAW264.7 cells ($P < 0.001$), while the ingredients with *m/z* values of 321.01 (digallic acid) and 787.19 (tetra-*O*-galloyl-β-D-glucose) didn't significantly inhibit the production of IL-6.

In summary, the ingredients with *m/z* values of 197.07 (EG), 483.22 (di-*O*-galloyl-β-D-glucose) and 635.30 (tri-*O*-galloyl-β-D-glucose) had stronger inhibitory effect on cytokines production compared with that in the ingredients with *m/z* values of 321.01 (digallic acid), 331.05 (mono-*O*-galloyl-β-D-glucose) and 787.19 (tetra-*O*-galloyl-β-D- glucose). Therefore, 3 categories of ingredients of *Galla chinensis*, the ingredients with *m/z* values of 197.07 (EG), 483.22 (di-*O*-galloyl-β-D-glucose) and 635.30 (tri-*O*-galloyl-β-D- glucose), were chosen to perform subsequent experiments with the concentration doses of 6.25, 12.5 and 25 μg/mL.

3.5. *The ingredients decreased the generation of ROS and increased TEER values in LPS- stimulated RAW264.7 cells*

To further investigate the antioxidant activity of the ingredients with *m/z* values of 197.07 (EG), 483.22 (di-*O*-galloyl-β-D-glucose) and 635.30 (tri-*O*-galloyl-β-D-glucose), the generation of ROS in the RAW264.7 cells was detected by flow cytometry. The experimental results showed that the intracellular ROS level in RAW264.7 cells was significantly increased when stimulated with LPS compared with the blank group ($P < 0.01$), while the ingredients with *m/z* values of 197.07 (EG), 483.22 (di-*O*-galloyl-β-D-glucose) and 635.30 (tri-*O*-galloyl-β-D-glucose) significantly inhibited the generation of ROS in LPS-activated RAW264.7 cells in a dose-dependent manner. In addition, the ingredient with *m/z* values of 197.07 (EG) markedly inhibited ROS generation in LPS-activated RAW264.7 cells ($P < 0.001$) (Fig. 6A and B).

To investigate the protective role of the ingredients with *m/z* values of 197.07 (EG), 483.22 (di-*O*-galloyl-β-D-glucose) and 635.30 (tri-*O*-galloyl-β-D-glucose) on the intestinal epithelium barrier *in vitro*, the intestinal physiological environment by culturing Caco-2 and RAW 246.7 cells together was simulated in a Transwell® plate (Fig. 6C). The results showed that LPS significantly decreased the resistance values of the monolayer (LPS 662.62 ± 80.30 Ω/cm^2 *vs.* control 1363.63 ± 37.64 Ω/cm^2; $P < 0.001$), whereas pre-treating with the ingredients with *m/z* values of 197.07 (EG), 483.22 (di-*O*-galloyl-β-D-glucose) and 635.30 (tri-*O*-galloyl-β-D-glucose) at 25 μg/mL observably inhibited this reducing effect ($P < 0.01$) (Fig. 6D).

The summary of the activity of each ingredient in *Galla chinensis* was showed in Table S2. According to the results, the ingredient with *m/z* 197.07 (EG) was found to be the major active ingredient. It not only played anti-inflammatory roles by reducing the expression of NO and cytokines, but also decreased the generation of ROS in LPS-activated RAW264.7 cells and maintained a proper intestinal epithelium barrier by inhibiting the inflammatory response of macrophages.

3.6. *Effects of EG on DSS-induced UC*

To assay whether the protective effects of EG in LPS-stimulated RAW 264.7 cells is related to clinic, the effects of EG on DSS-induced UC mice were tested. Mice drank water with 4% DSS for 18 days, and treated from the 7th day to assess the effect of EG on the mice of UC model as indicated in Fig. 7A. The body weight of mice in control group increased with the experiment period. In contrast loss of body weight in mice treated with DSS appeared from the sixth day. Similar to 5-ASA (100 mg/kg), a commonly used drug for treating UC, EG could robustly ameliorate the loss of body weight in a dose-dependent manner ($P < 0.001$), which was similar to the effects of CE at 250 mg/kg ($P < 0.001$) (Fig. 7B). Compared with the mice that only received water, the DAI score, an index of the severity of UC, was increased in DSS-exposed mice ($P < 0.001$), administration of 5-ASA (100 mg/kg), while EG at 150 mg/kg and 250 mg/kg or CE at 250 mg/kg evidently improved these symptoms ($P < 0.01$) (Fig. 7C). Moreover, the mice administrated with 5-ASA (100 mg/kg), EG at 150 mg/kg and 250 mg/kg, and CE at 250 mg/kg had significant alleviation of bloody diarrhea symptoms and the representative photographs of feces are presented in Fig. 7D.

In addition, due to the spleen/body weight ratio, spleen enlargement was observed in DSS-induced UC mice, and it was an organ index. Compared with the control group (DSS group 9.25 ± 2.05 mg/g *vs.* control group 2.53 ± 0.68 mg/g; $P < 0.001$), while 5-ASA at 100 mg/kg (3.59 ± 0.89 mg/g *vs.* DSS group 9.25 ± 2.05 mg/g), EG at 150 mg/kg therapy group (4.15 ± 1.42 mg/g *vs.* DSS group 9.25 ± 2.05 mg/g), and CE at 250 mg/kg (5.56 ± 0.62 mg/g *vs.* DSS group 9.25 ± 2.05 mg/g) could effectively reduce splenomegaly ($P < 0.01$) (Fig. 7E). Correspondingly, colon length clearly shortened after DSS administration compared with control mice (DSS group 6.78 ± 0.75 cm *vs.* control group 10.82 ± 1.29 cm; $P < 0.001$), another indicator for the UC severity of intestinal inflammation. 5-ASA at 100 mg/kg (5-ASA group 10.82 ± 1.29 cm *vs.* DSS group 6.78 ± 0.75 cm; $P < 0.001$), EG at 150 mg/kg (EG group 9.58 ± 1.34 cm *vs.* DSS group 6.78 ± 0.75 cm; $P < 0.001$) and CE at 250 mg/kg (CE group 8.58 ± 1.12 cm *vs.* DSS group 6.78 ± 0.75 cm; $P < 0.001$) could obviously alleviate the colon shortening induced by DSS (Fig. 7F and G). From these results, EG at 150 mg/kg and 5-ASA at 100 mg/kg showed comparable therapeutic effects in DSS-induced UC mice. Additionally, EG showed better therapeutic effect than CE at the same dose.

3.7. *EG ameliorated histopathological symptoms in DSS-induced UC*

The effects of EG on DSS-induced UC were evaluated by histopathological analysis. The acute phase of UC shows a series of histological features such as edema, mucosal erosions, crypt shortening, and

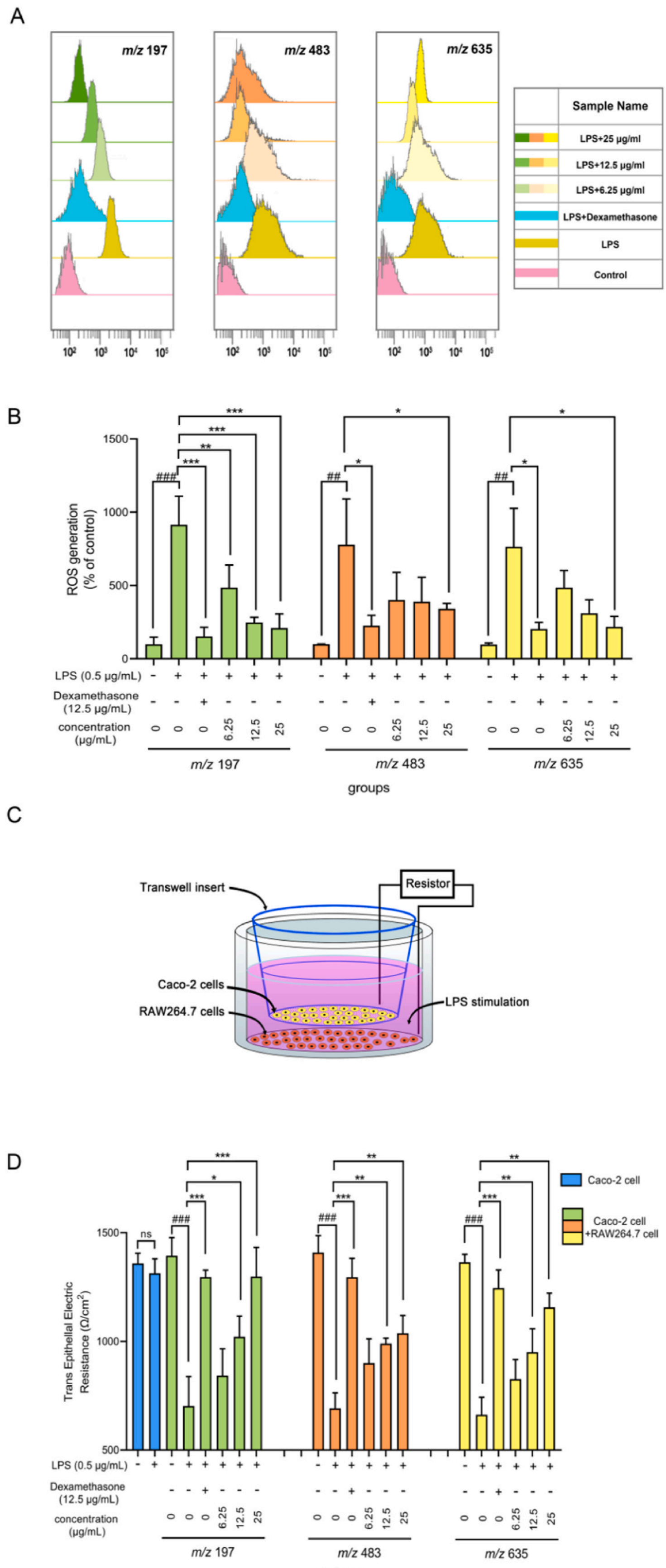

Fig. 6. The ingredients decreased the generation of ROS and increased TEER values in LPS-stimulated RAW264.7 cells. (A, B) The ROS generations induced by LPS in the RAW264.7 cell was calculated with the flow cytometry. (C) Schematic diagram of the testing process of TEER. (D) TEER of various treatments of the Caco-2 monolayer. The values are expressed as mean ± SD, n = 6. Values of $^{\#}P < 0.05$, $^{\#\#}P < 0.01$ and $^{\#\#\#}P < 0.001$ *vs.* the blank group. Values of $^{*}P < 0.05$, $^{**}P < 0.01$ and $^{***}P < 0.001$ *vs.* the model group.

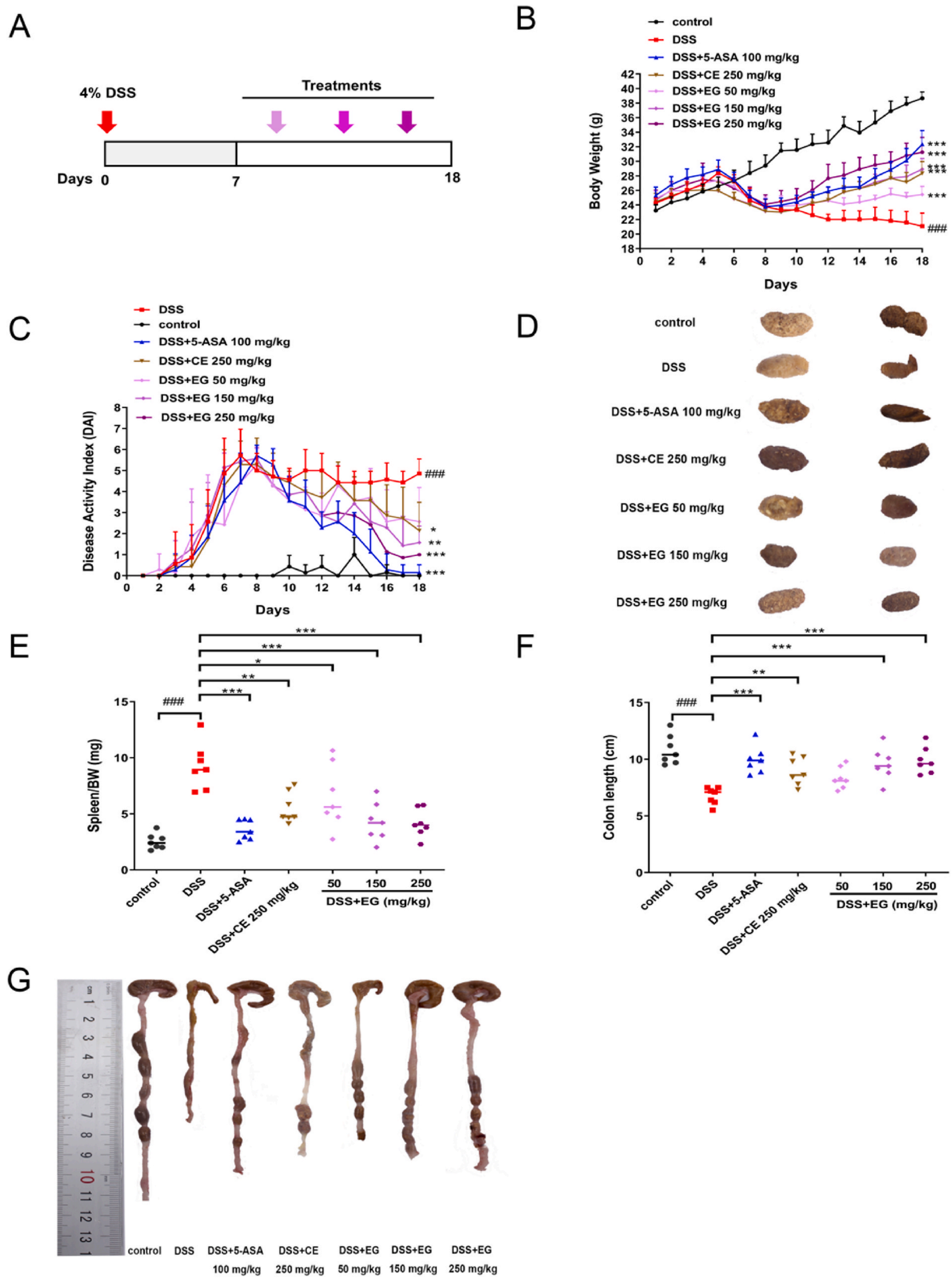

Fig. 7. EG prevented DSS-induced UC. (A) Mice were exposed to drinking water containing 4% DSS for 7 days, mice were given an intragastric. Administration of vehicle control, 5-ASA (100 mg/kg), CE (250 mg/kg) and EG (50 mg/kg, 150 mg/kg and 250 mg/kg) from day 7 to day 18. (B) Body weight curves of mice. (C) Daily DAI measurement. (D) Representative photographs of feces. (E) Spleen/BW ratio measured at the end of experiment. (F) colon length. (G) Representative photographs of colons. The values are expressed as mean ± SD, n = 7. Values of $^{\#}P < 0.05$, $^{\#\#}P < 0.01$ and $^{\#\#\#}P < 0.001$ *vs.* the blank group. Values of $^{*}P < 0.05$, $^{**}P < 0.01$ and $^{***}P < 0.001$ *vs.* the model group.

infiltration of inflammatory cells in the mucosa and lamina propria. As shown in Fig. 8A, compared with the control group, mice treated with DSS exhibited serious injuries of colon mucosa, a marked decrease in the number of crypts, strong epithelial erosion, loss of histological structures, and pronounced infiltration of inflammatory cells. While mice with UC treated with EG at 150 mg/kg and 250 mg/kg exhibited histological changes in pathological inflammation comparable to 5-ASA (100 mg/kg). Yet mice with UC treated with EG at 50 mg/kg had severe inflammatory infiltration of the tissue. Administration with 5-ASA (100 mg/kg), EG at 150 mg/kg and 250 mg/kg or EC at 250 mg/kg also decreased the histological scores ($P < 0.05$), while EG reduced histopathological score more obviously when EG and CE were in the same dose (Fig. 8B), indicating that EG at 150 mg/kg and 250 mg/kg exerted significant protective effects on intestinal damage in DSS-induced UC mice. Consequently, effects of EG on mice with UC were studied in the following.

3.8. EG regulated enzymes involved in oxidative stress responses and cytokines in DSS- induced UC

SOD and MDA are important signs of oxidative stress in colitis. MPO is an important marker of neutrophil inflammatory infiltration. As shown in Fig. 9A, DSS induced a significant decrease in SOD activity in the colon compared with the control group (DSS group 15.39 ± 1.96 U/mgprot *vs.* control group 30.19 ± 0.54 U/mgprot; $P < 0.001$). In contrast, treatment with EG at 250 mg/kg significantly increased the SOD activity (EG group 22.87 ± 1.06 U/mgprot *vs.* DSS group 15.39 ± 1.96 U/mgprot; $P < 0.001$). MDA and MPO activities increased after DSS

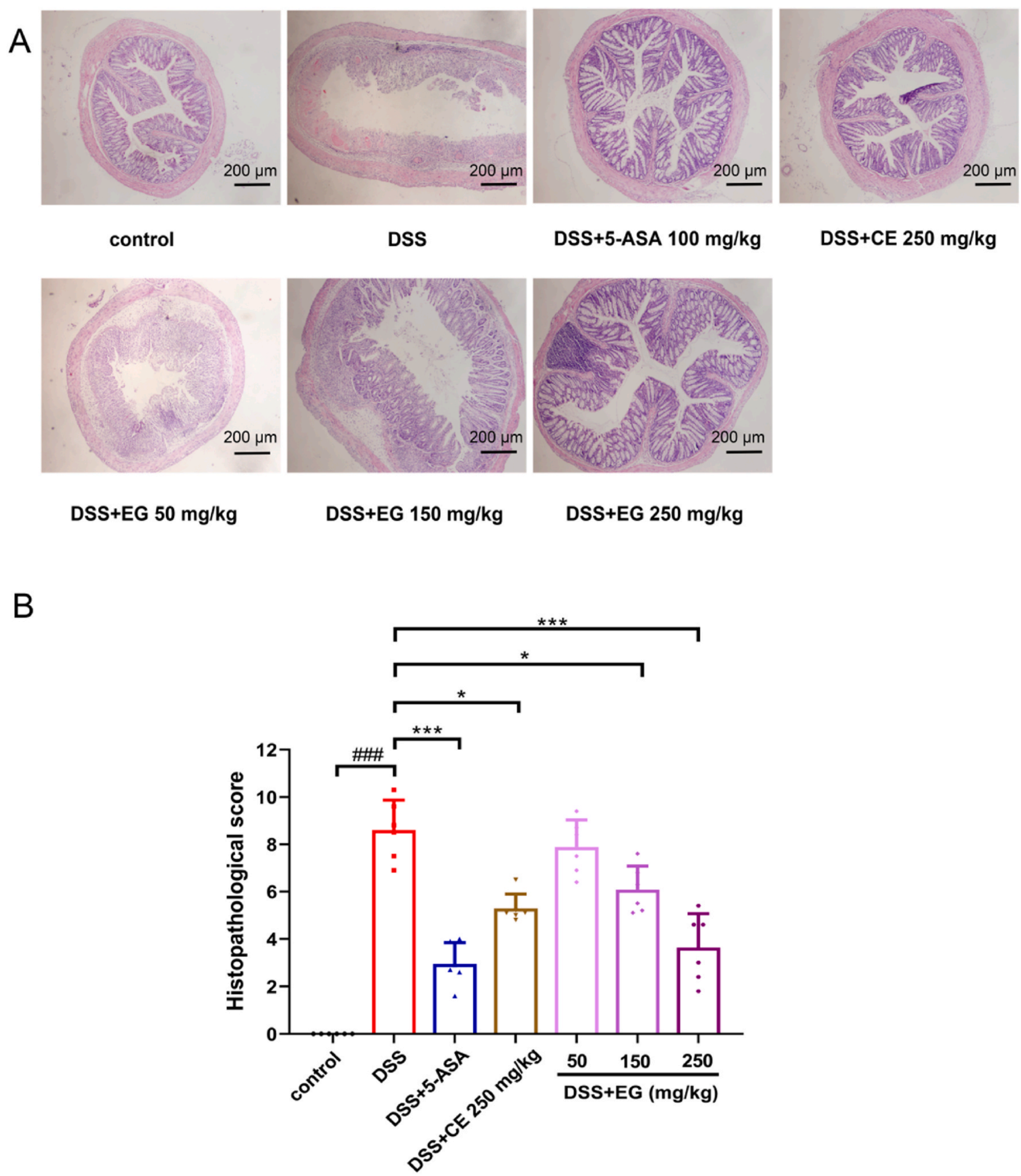

Fig. 8. EG relieved histopathological symptoms in DSS-induced UC. Colons were removed and sectioned followed by H&E staining. Representative sections (A) and histopathological score (B) are displayed. Scalebar = 200 μm. The values are expressed as mean ± SD, n = 6. Values of $^{\#}P < 0.05$, $^{\#\#}P < 0.01$ and $^{\#\#\#}P < 0.001$ *vs.* the blank group. Values of $^{*}P < 0.05$, $^{**}P < 0.01$ and $^{***}P < 0.001$ *vs.* the model group.

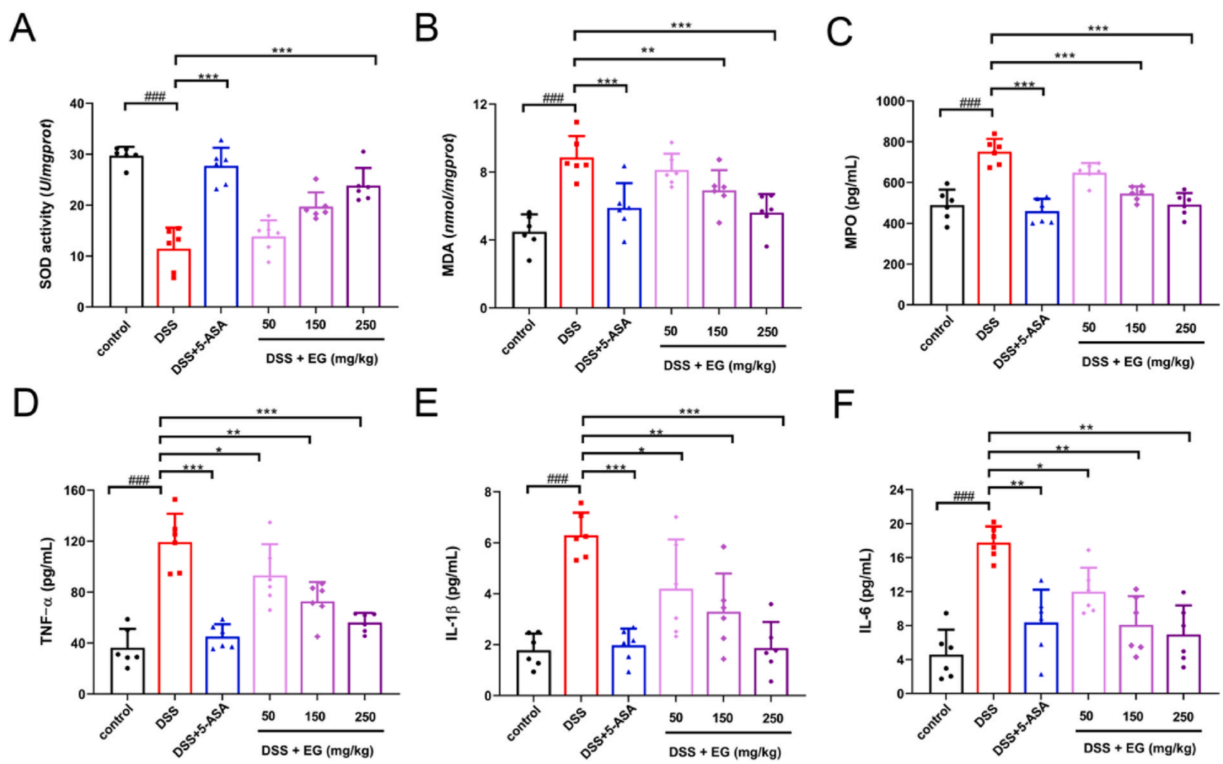

Fig. 9. EG regulated enzymes involved in oxidative stress responses and cytokines in DSS-induced UC. Colonic levels of SOD (A), MDA (B) and MPO (C) were examined by chemical chromatometry. Serum levels of proinflammatory cytokine including TNF-α (A), IL-1β (B) and IL-6 (C) were measured with ELISA. The values are expressed as mean ± SD, n = 6. Values of $^{\#}P < 0.05$, $^{\#\#}P < 0.01$ and $^{\#\#\#}P < 0.001$ *vs.* the blank group. Values of $^{*}P < 0.05$, $^{**}P < 0.01$ and $^{***}P < 0.001$ *vs.* the model group.

treatment ($P < 0.001$). After treatment with EG at 250 mg/kg, the activities of MDA and MPO were decreased to 1.5 times approximately (EG group 5.28 ± 0.34 pg/mL *vs.* DSS group 8.14 ± 0.56 pg/mL; $P < 0.001$) and 1.6 times approximately (EG group 451.59 ± 56.95 nmol/mgprot *vs.* DSS group 720.16 ± 47.74 nmol/mgprot; $P < 0.001$). The activities of MDA and MPO were significantly different from that of DSS group after EG treatment, and there was a dose dependence. (Fig. 9B and C).

UC is characterized by production of a wide range of inflammatory cytokines, including TNF-α, IL-1β and IL-6. The serum levels of inflammatory cytokines were determined by ELISA. The results demonstrated that serum levels of TNF-α, IL-1β and IL-6 in mice treated with DSS were substantially increased ($P < 0.001$). EG at 250 mg/kg group has the highest therapeutic effect, with TNF-α reduced to 1.8 times approximately (EG group 59.24 ± 6.55 pg/mL *vs.* DSS group 108.34 ± 16.06 pg/mL; $P < 0.001$). IL-1β was able to be reduced to 3 times approximately (EG group 1.93 ± 0.41 pg/mL *vs.* DSS group 5.78 ± 0.43 pg/mL; $P < 0.001$), and IL-6 levels was able to be decreased to 3.6 times approximately (EG group 4.65 ± 3.60 pg/mL *vs.* DSS group 16.78 ± 1.43 pg/mL; $P < 0.001$). (Fig. 9D, E and F).

The histopathological observation and the data of cytokines and antioxidant indexes from *in vivo* experiments attested that EG could reduce the severity of UC, which was consistent with the *in vitro* results. Subsequently, the mechanism of EG in treating UC was further explored according to the prediction by the systems pharmacology approach.

3.9. EG regulated the transcript expression levels of cytokines and oxidation markers in DSS-induced UC

To determine the effects of EG on DSS-induced UC, mRNA levels of the proinflammatory cytokines and Oxidation markers in the colons were measured by Q-PCR. As shown in Fig. 10, DSS treatment significantly upregulated proinflammatory cytokines, including TNF-α, IL-1β and IL-6 at the mRNA level ($P < 0.001$) (Fig. 10A, B and C), while treatment with EG at 150 mg/kg and 250 mg/kg or 5-ASA (100 mg/kg) almost completely suppressed the upregulation. In addition, heme oxygenase-1 (HO-1) at the mRNA level was significantly lower in the colons of DSS-induced mice than in healthy controls ($P < 0.05$), however, treatment with EG at 150 mg/kg and 250 mg/kg or 5-ASA (100 mg/kg) increased the expression of HO-1 ($P < 0.05$) (Fig. 10D). NQO1 at the mRNA level was substantially downregulated in mice with DSS-induced UC and treatment with EG was able to increase NQO1 expression, although not significant enough. (Fig. 10E). In contrast, inducible nitric oxide synthase (iNOS) at the mRNA level was markedly upregulated in UC mice ($P < 0.05$) (Fig. 10F). CAT and SOD_2 at the mRNA level were substantially down-regulated in mice with DSS-induced UC ($P < 0.05$) (Fig. 10G and H). However, mice treated with either EG at 150 mg/kg and 250 mg/kg or 5-ASA (100 mg/kg) significantly antagonized DSS-induced effects. Data from mRNA regulation demonstrated that EG exerted therapeutic effects on DSS-induced UC through anti-inflammatory and anti-oxidation.

3.10. EG decreased inflammation and oxidative stress via activation of Nrf2 and NF-κB to protect the intestinal mucosal barrier

Our findings described above supported a potent negative regulatory role of EG in suppressing inflammatory responses and oxidative stress and protective effect of intestinal mucosal barrier. To further validate the underlying mechanism behind the protective role of EG, the associated signaling pathways accounting for therapeutic effects were determined. Occludin and Claudin-1 are important TJs, which are necessary to maintain the integrity of the normal barrier. As shown in Fig. 11A and B, the expression of Occludin and Claudin-1 were found to be significantly reduced in DSS-induced mice ($P < 0.01$). By contrast, the expression of Occludin and Claudin-1 were considerably increased in the treatment group, along with significant differences between 5-ASA (100 mg/kg) group and EG at dose of 250 mg/kg treatment group ($P < 0.05$).

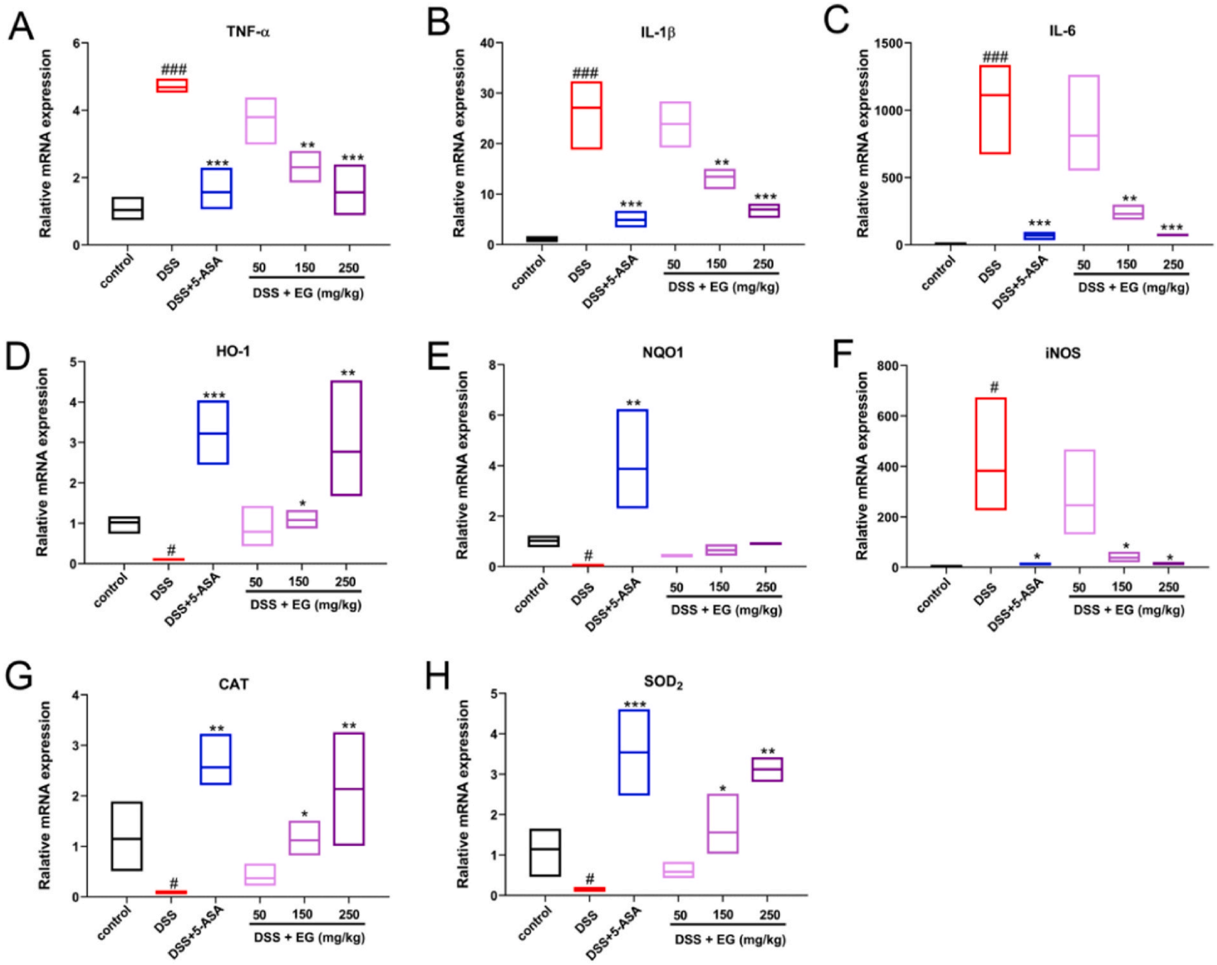

Fig. 10. EG regulated the transcript expression levels of cytokines and oxidation markers in DSS-induced UC. mRNA expression levels of the proinflammatory genes including TNF-α (A), IL-1β (B) and IL-6 (C) as well as oxidative stress-related genes including HO-1 (D), NQO1 (E), iNOS (F), CAT (G) and SOD_2 (H) in colonic tissues were determined by real-time PCR. The housekeeping gene β-actin was used as a loading control. The values are expressed as mean ± SD. Values of $^{\#}P < 0.05$, $^{\#\#}P < 0.01$ and $^{\#\#\#}P < 0.001$ *vs.* the blank group. Values of $^{*}P < 0.05$, $^{**}P < 0.01$ and $^{***}P < 0.001$ *vs.* the model group.

The results of systems pharmacology showed that NF-κB is an important pathway of UC, and played a critical role in inflammation. Then the effects of EG on NF-κB activation was examined in colon tissues. As a key regulator of detoxification, Nrf2 is responsible for the transcriptional activation of genes encoding for antioxidant enzymes. Results showed that mice with DSS-induced UC exhibited substantially increased expression of P-IKKβ and P–P65 compared with the control group ($P < 0.01$), however, the expression of these proteins was markedly elevated following treatment with EG at dose of 250 mg/kg or 5-ASA (100 mg/kg) ($P < 0.05$) (Fig. 11C, E). Meanwhile, the unphosphorylated IKKβ and NF-κB p65 did not change (Fig. 11D, F). In order to evaluate the antioxidant activity of EG, the levels of oxidative stress-related proteins were examined. DSS-induced UC mice resulted in an increased expression of Keap1 in colon tissues, while treatment with EG at dose of 250 mg/kg or 5-ASA (100 mg/kg) displayed a significant reduction of Keap1 protein ($P < 0.05$) (Fig. 11H). Furthermore, protein expression of HO-1 and Nrf2 in mice treated with DSS was significantly decreased as a result of UC ($P < 0.05$), but EG at dose of 250 mg/kg or 5-ASA (100 mg/kg) was capable of restoring or even further enhancing the expression of Nrf2 and HO-1 ($P < 0.05$) (Fig. 11G, I). The data of protein expression demonstrated that EG exerted therapeutic effects in experimental UC through activation of Nrf2 and degradation of Keap1 to protect the intestinal mucosal barrier.

4. Discussion

EG regulated proteins related to the NF-κB and Nrf2 pathway to protect intestinal mucosa in UC (Fig. 12). EG itself not only played an antioxidant role, but also protected the integrity of intestinal mucosa through negative regulation of NF-κB related proteins and positive regulation of Nrf2 pathway related proteins in UC. On the one hand, p-IKKβ phosphorylated P65 was activated and entered the nucleus to produce a large number of inflammatory cytokines such as IL-1β, IL-6, TNF-α, which lead to destruction of intestinal mucosa epithelium; on the other hand, Nrf2 dissociated from Keap1 and translocated to the nucleus under oxidative stress. Free Nrf2 combined with sMaf protein to form antioxidant element ARE, and then its target genes HO-1/NQO1 were activated to alleviate oxidative damage.

The discovery of a group of ingredients that can basically represent the overall efficacy of the original prescription from the many ingredients of traditional Chinese medicine can be used as a clue to the discovery of the lead compound. At present, there were not many studies on finding lead compounds in natural products through systems pharmacology coupled with equivalent components group technology. In this research, a stepwise tracking strategy of systems pharmacology coupled with equivalent components group technology based on mass spectrometry guided preparative chromatography was adopted to screen the active ingredients in *Galla Chinensis* and predict their corresponding targets. We found that each ingredient of *Galla chinensis*

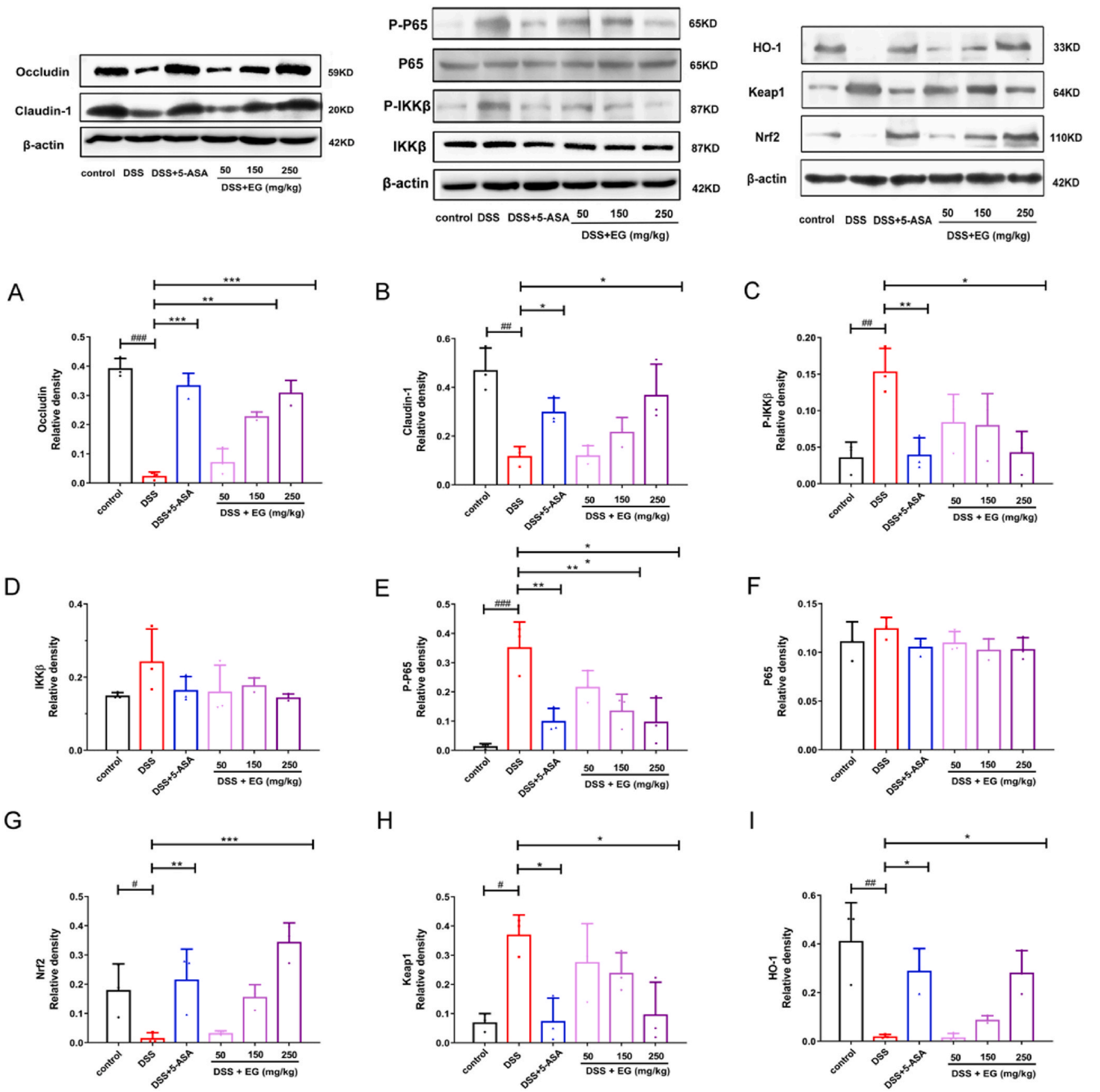

Fig. 11. EG decreased inflammation and oxidative stress via activation of Nrf2 and NF-κB to protect the intestinal mucosal barrier. The expression of Occludin, Claudin-1, IKKβ, P-IKKβ, NF-κB p65, P–P65, Nrf2, HO-1 and Keap1 in colonic tissues were assessed with western blotting. Representative data is displayed and the relative protein intensity of Occludin, Claudin-1, IKKβ, P-IKKβ, NF-κB p65, P–P65, Nrf2, HO-1 and Keap1 was normalized to β-actin (A–I). The values are expressed as mean ± SD. Values of $^{\#}P < 0.05$, $^{\#\#}P < 0.01$ and $^{\#\#\#}P < 0.001$ *vs.* the blank group. Values of $^{*}P < 0.05$, $^{**}P < 0.01$ and $^{***}P < 0.001$ *vs.* the model group.

corresponded to multiple targets and multiple pathways. Especially, EG, di-*O*-galloyl-β-D-glucose and tri-*O*-galloyl-β-D-glucose acted on a wealth of targets such as TNF, SOD_2, CAT, NQO1, IL-6 and IL-1β were closely related to the UC process. In addition, NF-κB and Nrf2-Keap1 signaling pathways played the highest role and important target points.

Though the exact etiologies of UC were yet to be known, the fact that imbalance between Nrf2 and NF-κB pathways was associated with UC was increasingly recognized (Pan et al., 2012). UC was resulted from oxidative damage caused by excessive free radicals. Nrf2-mediated transcription was inducible under xenobiotic and oxidative stresses (Aviello and Knaus, 2017) and previous findings had revealed that the transcription factor Nrf2 could regulate the expression of phase II detoxifying enzymes such as NQO1, glutathione peroxidase, ferritin, HO-1 and antioxidant genes, which were crucial for protecting cells from various injuries, thus influencing disease progression (Sjölund et al., 2015). In addition, previous findings have demonstrated that mucosal inflammation resulted in profound shifts in the intestinal epithelial homeostasis, which may increase barrier permeability through influencing intercellular adhesions. An alteration in the phosphorylation pattern caused by inflammation or elevated ROS, can alter the protein–protein interactions of Occludin with Zona occludin-1 (ZO-1) and Claudins, thus changing the membrane integrity (Kevil et al., 2015). In this research, we found that EG can treat mice with UC and discussed the effect of EG in the treatment of UC from the three

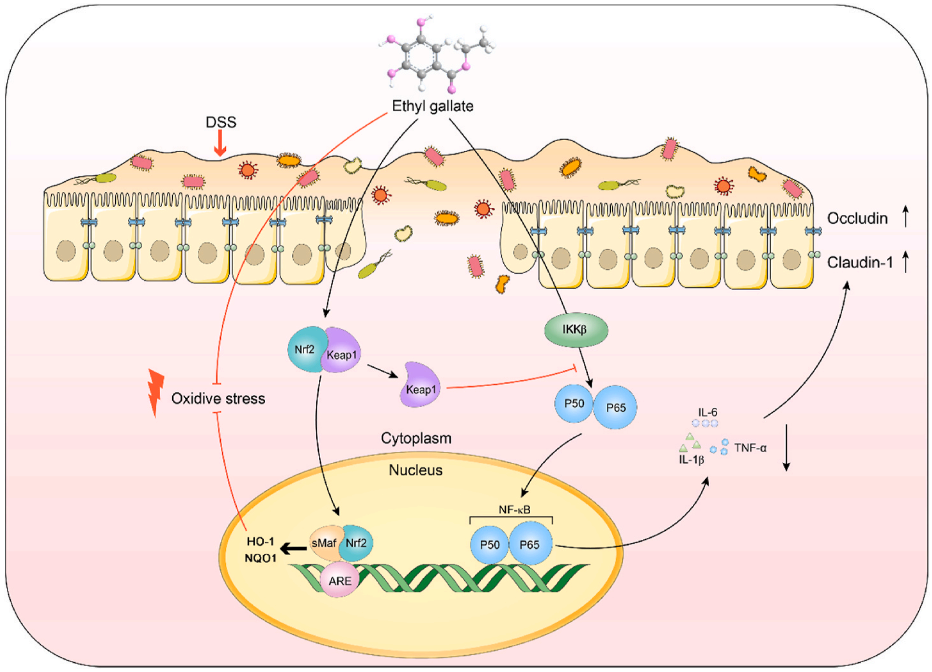

Fig. 12. EG regulated proteins related to the NF-κB and Nrf2 pathway to protect intestinal mucosa in UC.

aspects of inflammation, oxidative stress, and intestinal mucosal integrity. EG inhibited the expression of IL-6, IL-1β and TNF-α and increased the expression of Nrf2, HO-1, NQO1, CAT, SOD_2, Occludin and Claudin-1 to alleviate inflammatory symptoms in UC mice. Moreover, we found that its mechanism of action was achieved through protected the integrity of intestinal mucosa through negative regulation of NF-κB pathway and positive regulation of Nrf2 pathway.

5. Conclusion

In conclusion, a stepwise tracking strategy of systems pharmacology coupled with equivalent components group technology based on mass spectrometry guided preparative chromatography was adopted to screen the lead compound and active ingredient in *Galla chinensis*. EG was screened out as the lead compound and active ingredient of *Galla chinensis* and it was discovered for the first time that EG could treat mice with UC. In addition, we discussed the effect of EG in the treatment of UC from the three aspects of inflammation, oxidative stress, and intestinal mucosal integrity, and found that its mechanism of action was achieved through the Nrf2 and NF-κB pathways. This research not only found the lead compound of *Galla Chinensis* for UC treatment and determined the possible mechanism, but also provided valuable references for finding lead compounds from natural products by systems pharmacology coupled with equivalent components group technology.

Authors' contribution

All authors contributed extensively to the work presented in this paper. Zhongying Wang and Bo Han designed the experiments, analyzed data, prepared figures and wrote up the manuscript. Rui Xue and Yunyun Qi provided helps for experiments and data analysis. Rui Xue, Mengying Lv and Wei Yu reviewed the manuscript. Wen Chen, Zhiyong Xie, Xinjun Wang and Xing Tian provided helps for the laboratory technique. The manuscript has been reviewed and approved by all authors.

Ethics approval and informed consent

All animal procedures were performed in accordance with the Guidelines for Care and Use of Laboratory Animals of Shihezi University and approved by the Animal Ethics Committee of the First Affiliated Hospital of the Medical College, Shihezi University.

CRediT authorship contribution statement

Zhongying Wang: Validation, Investigation, Formal analysis, Resources, Data curation, Writing – original draft. **Rui Xue:** Resources, Data curation, Writing – original draft, Writing – review & editing. **Mengying Lv:** Writing – review & editing, Funding acquisition. **Yunyun Qi:** Formal analysis, Data curation. **Wei Yu:** Writing – review & editing, Visualization. **Zhiyong Xie:** Supervision, Project administration. **Wen Chen:** Supervision, Project administration. **Xinjun Wang:** Supervision, Project administration. **Xing Tian:** Supervision, Project administration. **Bo Han:** Conceptualization, Methodology, Software, Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jep.2021.114533.

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